GCSE

Rewarding Learning

Chief Examiner's Report

Digital Technology



Summer Series 2018

Foreword

This booklet outlines the performance of candidates in all aspects of CCEA's General Certificate of Secondary Education (GCSE) in Digital Technology for this series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCSE DIGITAL TECHNOLOGY

Chief Examiner's Report

Assessment Unit 1 Digital Technology

Q1 These multiple choice questions were well attempted by the majority of candidates and many candidates achieved full marks. Part (g) about legislation and the role of the Information commissioner proved challenging, with many candidates selecting D rather than B as their response. It is worth noting that legislation is an important aspect of the Digital Technology curriculum and candidates should be familiar with the laws and the roles of individuals associated with legislation. Part (h) was less well answered with many candidates selecting D instead of B.

Q2 This was a basic database question worth 5 marks. Candidates were provided with a diagram of the database structure. Candidates should be able to understand the concept of key fields and the use of common fields to link tables. They should also be able to explain the different types of relationships. Most candidates were able to select the correct data type for Joining Fee, however a lesser number selected Data/Date Time for the DateOfBirth field. Again it is expected that candidates would be able to select appropriate data types for fields in context. Part (c) was not well attempted. Candidates did not articulate well the purpose of the MemberNewsletter table. Incorrect answers 'included to create a relationship between all three tables' 'and it is to keep the tables linked'. Candidates should understand the significance of this table and were expected to explain that the table kept the details of members and the newsletters they received. This type of table is common in database systems so, candidates should become familiar with the role and function of such a table. The definitions for primary key and the relationship type were well answered.

Q3 This question related to 'big data'. Overall the question was not well answered with only a small number of candidates achieving marks in the top band. In Part (a) candidates' answers were brief and did not include the fact that data analytics included the application of algorithms or programs to identify patterns and trends. In Part (b) many candidates re-stated the stem and did not adequately explain the meaning of the terms volume, velocity and variety as they relate to 'big data'. Some candidates incorrectly referenced velocity as the speed of the data transmission. Marks were made available if candidates were able to give suitable examples. Again, only a small number of candidates were able to give examples of the use of 'big data' and how it might be generated. Candidates should be encouraged to examine the data they generate as individuals, this will provide practical and memorable examples for contextual use. In addition, candidates should be reminded that such data cannot be processed using traditional data processing applications.

Q4 This question on graphics was well attempted by candidates. They were able to demonstrate a good understanding of the different graphic file formats and their relative properties. Part (a)(i) was well answered with many candidates accessing both marks. Acceptable answers included reference to 'pixels each having their own colour', 'being the smallest unit of an image' and 'when combined together form an image'. A typical correct answer gaining two marks was 'a pixel is the smallest unit of an image that can be edited'. Part (a)(ii) was also well attempted with many candidates achieving at least one mark. Whilst 'resolution is a measure of the quality of an image' was acceptable for 1 mark, reference to how clear the image is was not. Many candidates further expanded their answer to include the number of pixels per inch or include a comparison such as the more pixels the greater the quality of

the image. In Part 4 (b)(i) many candidates correctly expanded the acronyms. PNG was the most challenging of the two and some candidates still use picture instead of photographic with regard to JPEG. Candidates were able to refer to compression when answering Part (b)(ii), there was good use of technical terminology here and the best answers referred to lossy compression. Part 4 (b)(iii) was also well answered with a significant number of candidates achieving the 1 mark available. Typical correct answers were 'vector based graphics used mathematical equations, lines and curves'.

Q5 Candidates demonstrated a good understanding of the computer specification and a significant number were able to achieve at least 2 marks in Part (a)(i). It is worth noting that 'dual' was not acceptable as an answer for the number of cores.

Part (a)(ii) was less well answered. Candidates were expected to make a simple comparison, for example the greater the clock speed the greater the processing power or clock speed influences the speed at which the cpu executes commands. Such responses attracted 1 mark. Candidates should be aware of the relative performance factors within the computer system.

In Part (a)(iii) candidates were expected to recognise that the computer had a solid state drive and were credited for doing so. As part of the specification they should also be aware of the effectiveness of an SSD. This would have attracted the second mark. Some candidates referred to the storage capacity of the device, but this was not credited in this case as it was stated in the stem.

Part (b) relating to system software was not well answered. A significant number of candidates achieved 1 or 2 marks for stating resources managed by the system software. However, very few achieved full marks in the question about allocating memory. Candidates were required to know that data must be moved from the hard disk to a free space in RAM and transferred out of RAM when no longer required.

Part (c) about processing was well attempted by candidates across the range, many achieving 3 or 4 marks.

Q6 In Part (a), candidates were able to describe the purpose of levels of access e.g. to prevent unauthorised access. However, many were unable to give concrete examples of how the data is protected using levels of access. Candidates should be able to state some of the levels of access and describe, in simple terms, how these can contribute to protecting data. Many answers referred to hacking which was incorrect.

The QWC question relating to a star network in Part (b) was well answered by candidates, many of whom scored in Level 2. Candidates in the top band provided a description of the topology, the role of the server and the impact of node failure or server failure.

Q7 The spreadsheet question was well answered, particularly in Parts (a) and (b). Part (c) relating to vlookup (a new term on the specification) was less well answered. Very few candidates were able to articulate the advantage of using a vlookup table or an absolute cell reference. A practical understanding of how the vlookup works is required by the specification and because there is no practical assessment of spreadsheets in the Controlled Assessment, it is vital that candidates are exposed to solid exemplification of this and other complex areas such as macros.

The question in Part (d) which referred to conditional formatting, was also poorly answered and a minority of candidates achieved 1 mark for this.

Q8 Most candidates were able to distinguish the difference between ftp and http in Part (a). Parts (b) and (c) were also well answered. However in Part (b) on occasions candidates gave two answers relating to the same thing. For example, 'blocks unknown data from entering and controls what enters and leaves' would have been allocated 1 mark together. Candidates distinguished well between the differing types of cyber-crime, many achieving a full 3 marks for Part (c). Part (d) required candidates to explain how a worm works to disrupt a system. Many candidates achieved 1 or 2 marks in this part of the question. Most responses acknowledge that a worm slows a computer down and replicates itself. Only a minority of candidates recognised that no host is required or provided an example of what a worm would actually do to a system; either of these points would have achieved an extra mark.

Q9 Parts (a) and (b) related to the use of cloud computing. There were some very good answers provided for both and many candidates achieved a full 4 marks here. However, it is worth noting that candidates who mentioned 'can be accessed from anywhere' but did not mention the need for an Internet connection, were not awarded any marks. Also, in Part (b) no marks were awarded for answers relating to downtime for updates.

Part (c) related to online distribution of games. Candidate responses indicated a good awareness of the benefits for both customer and vendor.

Q10 Part (a) was not as well answered as expected. Common incorrect answers included Global Positioning Satellite (as opposed to System). In Part (b) candidates demonstrated a good knowledge of privacy and online safety. A high proportion gained full marks for the question about GPS and privacy, indicating that the message about online safety is getting across to candidates in this age group. Part (c) was also well answered. Top answers included reference to posture for back pain and ergonomic furniture for RSI. A significant minority still incorrectly referred to comfortable chairs and gained no marks in that response.

Q11 Part (a) was well attempted with many candidates gaining a full 2 marks for correctly naming two devices. Some candidates lost 1 mark by responding using phone, this was considered too generic and so attracted no marks.

Part (b) was misinterpreted by a minority of candidates who responded using advantages to the employee (as opposed to the employer). Candidates should be encouraged to read the questions very carefully and underline important requirements for answering purposes.

Q12 This question asked candidates to match terms to their definition and was well received. Many candidates achieved at least 3 of the 4 marks available. The definitions for router and Bluetooth were most commonly correct, whilst the definitions for defragmenting and program counter were incorrect. In some cases, candidates chose the Memory Data Register for the definition of the program counter.

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